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Bihar Engineering University, Patna

B.Tech 8th Semester Exam., 2021

Code: 061810	***Subject: MULTIMEDIA TECHNOLOGY AND APPLICATION***	***Time: 03 Hours***
Semester: VIII		***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Answer any seven of the following questions : ***[2 x 7 = 14]***

(a) Define the term 'interactive media'.

(b) List all the basic components of multimedia system.

(c) What is the difference between analog video and digital video?

(d) List the names of various image formats.

(e) Define the term MIDI.

(f) Define the process of video capturing.

(g) What is the advantage of indexing using Quad tree?

(h) What are the differences between SGML and HTML?

(i) Explain the term ODA.

(j) Define the term 'content management'.

Q.2 Attempt all parts : ***[7+7=14]***

(a) What are the hardware and software requirements for multimedia computer? Explain it.

(b) A digital video is to be compressed using the MPEG-I standard. Assuming a frame sequence of : IBBPBBPBBPBBI..., and average compression ratio of 10 : 1 (I), 20 : 1 (P) and 50 : 1 (B), calculate the average bit rate that is generated by the encoder for NTSC digitization format.

Q.3 Attempt all parts : ***[7+7=14]***

(a) What are the multimedia authoring tools? Explain any three in brief.

(b) Explain QBIC and video content querying.

Q.4 ***[14]***

Explain the framework of content-based image retrieval (CBIR) system using color texture and shape features. Also state its applications.

Q.5 Attempt all parts : ***[7+7=14]***

(a) Suppose we view a decompressed 512 × 512 JPEG image but use only the color part of the stored image information, not the luminance part to decompress. What does the 512 × 512 color image look like? Assume JPEG is compressed using a 4 : 2 : 0 schemes.

(b) With the aid of a diagram, explain the working principle of PCM encoder.

Q.6 Attempt all parts : ***[7+7=14]***

(a) What are basic concepts of color? Explain it.

(b) Explain Hypertext and its features.

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Q.7 Attempt all parts : ***[7+7=14]***

(a) With the aid of a diagram, explain the five main stages associated with baseline mode of operation of JPEG and give a brief description of the role of each stage.

(b) Explain the terms 'storage models' and 'access techniques' in brief.

Q.8 Attempt all parts : ***[7+7=14]***

(a) Discuss the design principles of content management.

(b) Describe the principle of H.261 in detail.

Q.9 Attempt all parts : ***[7+7=14]***

(a) An audio signal has a dynamic range of 40 dB. Determine the magnitude of quantization noise relative to the minimum signal amplitude if the quantizer uses (i) 5 bits and (ii) 8 bits. Which bit representation is acceptable for this audio signal?

(b) Explain multimedia archives and digital libraries with suitable examples.

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